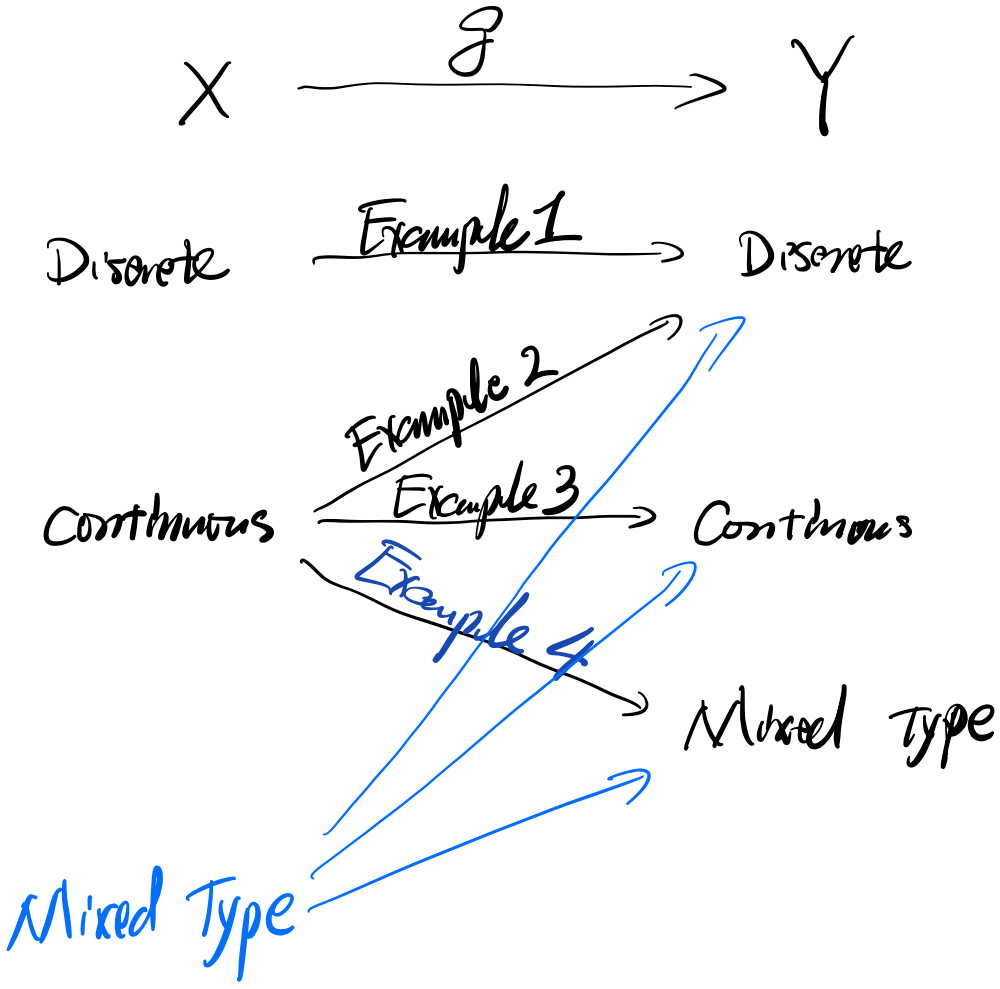
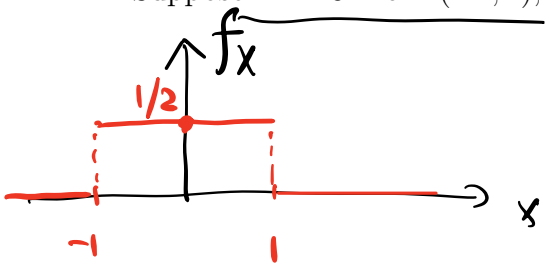
Functions of a Random Variable

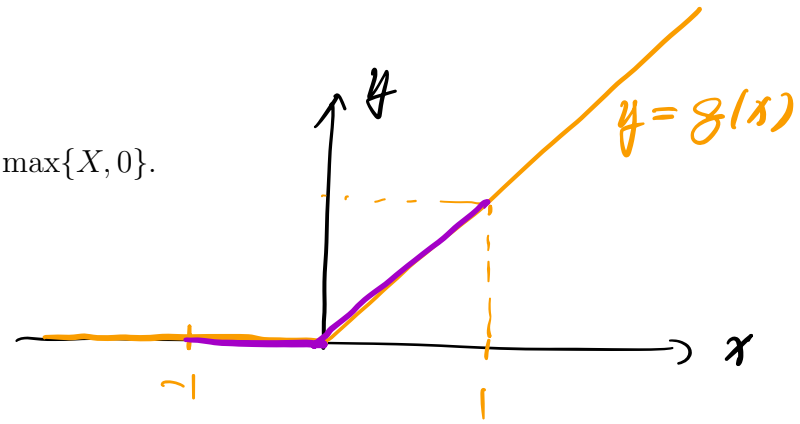
Question: If X is discrete (or continuous), must $Y = g(X)$ also be discrete (or continuous)?



Example 4 [Continuous to Mixed]

Suppose $X \sim \text{Uniform}(-1, 1)$, and $Y = \max\{X, 0\}$.





The CDF of Y :

$$F_Y(y) = P(Y \le y) = P(\max\{X, 0\} \le y)$$

Case 1 $y < 0$

Since $Y \ge 0$, $F_Y(y) = P(Y \le y) = 0$

Case 2. $0 \le y \le 1$

$$\max\{X, 0\} \le y \iff X \le y$$

$$\therefore F_Y(y) = P(X \le y) = \int_{-\infty}^{y} f_X(x)\,dx$$

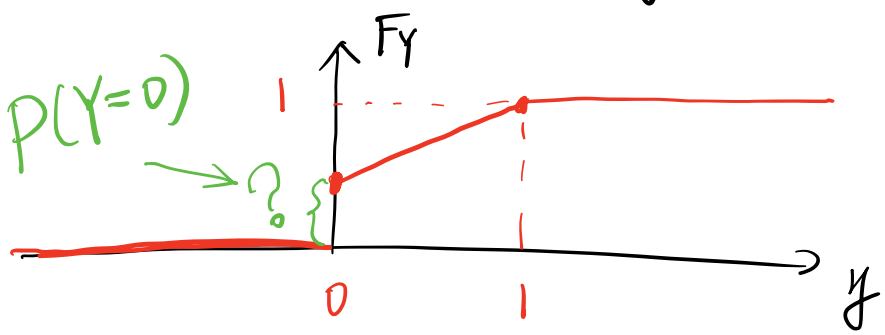
$$(= F_X(y)) \qquad = \int_{-1}^{y} \frac{1}{2}\,dx$$

$$= \frac{y+1}{2}$$

Case 3 : $y > 1$

Since $Y \le 1$, $F_Y(y) = P(Y \le y) = 1$

$$\therefore F_Y(y) = \begin{cases} 0 & , y < 0 \\ \frac{y+1}{2} & , 0 \le y \le 1 \\ 1 & , y > 1 \end{cases}$$

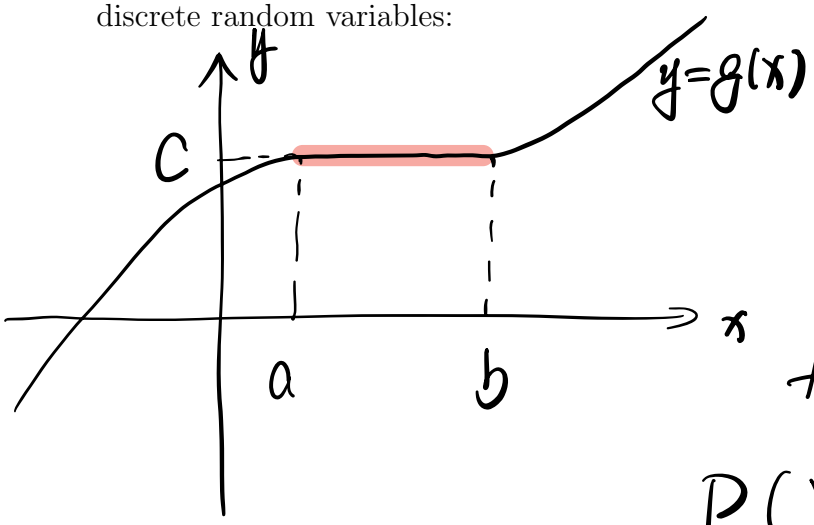


Y is of mixed type.

Functions: Continuous to Continuous

Let X be a continuous random variable and $g(u)$ is a continuous function of u such that $\frac{dg(u)}{du}$ **is not zero on any interval**, then $Y = g(X)$ is a continuous random variable.

Note: Having $\frac{dg(u)}{du} = 0$ for some values of u is acceptable, but $\frac{dg(u)}{du} = 0$ for all $u \in (a, b)$ is not allowed. This restrict avoids "flat regions" in the curve of $g(u)$ that may result in mixed or discrete random variables:



If $\frac{dg(u)}{du} = 0$ for $u \in (a, b)$.

then

$P(Y = c) = P(a \leq X \leq b)$

may not be zero. That is the probability mass can be assigned to a value of Y, and then Y is not a continuous RV.

Distributions of Monotonic functions of a Random Variable

When a random variable X is transformed by a monotonic function g, the probability distribution of the resulting random variable $Y = g(X)$ can be determined from the distribution of X using the cumulative distribution function (CDF) method.

Now suppose X is a continuous random variable with cumulative distribution function $F_X(x)$ and probability density function $f_X(x)$.

Let's consider a monotonic (strictly increasing or decreasing) function g.

$a < b \iff g(a) < g(b)$

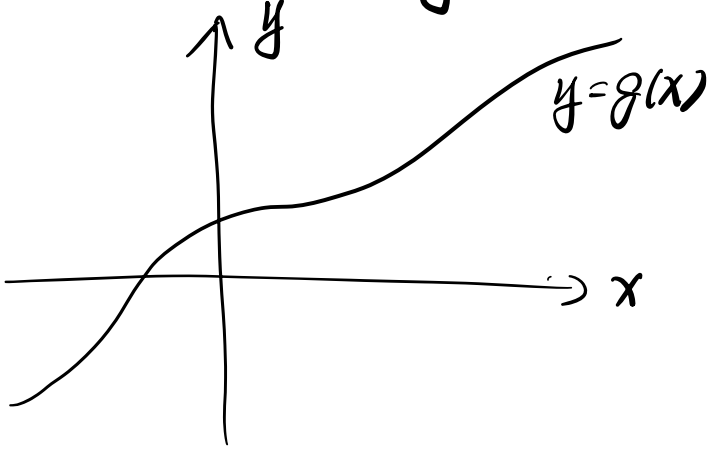
Objective: determine the distribution (PDF or CDF) of Y.

Suppose g is increasing function, then

$a \le b \iff g(a) \le g(b)$.

Assumption: ($\frac{dg(x)}{dx}$ is not zero over any interval)

and g is invertible, and g^{-1} is also increasing.



The CDF of Y:

$$F_Y(y) = P(Y \le y)$$

$$= P(g(X) \le y)$$

$$= P(X \le g^{-1}(y))$$

$$= F_X(g^{-1}(y))$$

That is,

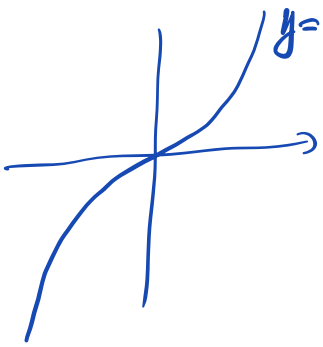
$$F_Y(y) = F_X(g^{-1}(y))$$

then the PDF is

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Example. $Y = X^3$ if $X \sim \text{Uniform}\,(-1, 1)$.

$y = g(x) = x^3$



$y = g(x)$ is strictly increasing over $x \in \mathbb{R}$.

$y = x^3 \iff x = y^{1/3} = g^{-1}(y)$

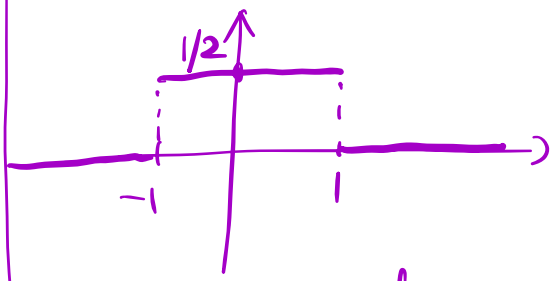
The CDF of Y:

$$F_Y(y) = F_X(g^{-1}(y))$$

$$= \begin{cases} 0 & , \sqrt[3]{y} < -1 \\ \frac{\sqrt[3]{y}+1}{2} & , -1 \le \sqrt[3]{y} \le 1 \\ 1 & , \sqrt[3]{y} > 1 \end{cases}$$

$$= \begin{cases} 0 & , y < -1 \\ \frac{\sqrt[3]{y}+1}{2} & , -1 \le y \le 1 \\ 1 & , y > 1 \end{cases}$$

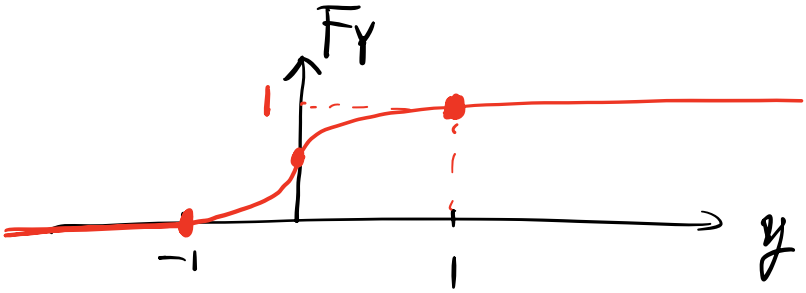
Recall: $X \sim \text{Uniform}\,(-1, 1)$



The CDF of X:

$$F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(x)\,dx$$

$$= \begin{cases} 0 & , x < -1 \\ \frac{x+1}{2} & , -1 \le x \le 1 \\ 1 & , x > 1 \end{cases}$$



Graph of the CDF $F_Y(y)$

Its PDF is

$$f_Y(y) = \frac{d F_Y(y)}{dy} = \begin{cases} 0, & y < -1 \\ \frac{1}{6 y^{2/3}}, & -1 < y < 1 \ \& \ y \neq 0 \\ 0, & y > 1 \end{cases}$$

Because $F_Y(y)$ is not differentiable at $y = 0$.

$$= \begin{cases} \frac{1}{6 y^{2/3}}, & -1 < y < 1 \ \& \ y \neq 0 \\ 0, & \text{otherwise} \end{cases}$$